

LAB: 02

Question 1

```
[mysql> Create database Freelance_Marketplace;  
Query OK, 1 row affected (0.04 sec)
```

```
[mysql> use Freelance_Marketplace;  
Database changed
```

```
mysql> CREATE TABLE CLIENT (  
  -> client_id VARCHAR(10) PRIMARY KEY,  
  -> full_name VARCHAR(100) NOT NULL,  
  -> email VARCHAR(100) UNIQUE NOT NULL,  
  -> phone VARCHAR(20) UNIQUE,  
  -> country VARCHAR(50) NOT NULL,  
  -> company_name VARCHAR(100),  
  -> avg_rating DECIMAL(3,2) DEFAULT 0 CHECK(avg_rating BETWEEN 0 AND 5),  
  -> created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
[  -> );  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE FREELANCER (  
  -> freelancer_id VARCHAR(10) PRIMARY KEY,  
  -> full_name VARCHAR(100) NOT NULL,  
  -> email VARCHAR(100) UNIQUE NOT NULL,  
  -> phone VARCHAR(20) UNIQUE,  
  -> country VARCHAR(50) NOT NULL,  
  -> bio TEXT,  
  -> hourly_rate DECIMAL(10,2) CHECK(hourly_rate > 0),  
  -> avg_rating DECIMAL(3,2) DEFAULT 0 CHECK(avg_rating BETWEEN 0 AND 5),  
  -> total_jobs_done INT DEFAULT 0 CHECK(total_jobs_done >= 0),  
  -> created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
[  -> );  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE SKILL (  
  -> skill_id VARCHAR(10) PRIMARY KEY,  
  -> skill_name VARCHAR(100) UNIQUE NOT NULL,  
  -> description VARCHAR(255)  
[  -> );  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE CATEGORY (  
  -> category_id VARCHAR(10) PRIMARY KEY,  
  -> category_name VARCHAR(100) UNIQUE NOT NULL,  
  -> description TEXT  
  -> );  
Query OK, 0 rows affected (0.01 sec)
```

```

mysql> CREATE TABLE JOB_POST (
  ->   job_id VARCHAR(10) PRIMARY KEY,
  ->   client_id VARCHAR(10) NOT NULL,
  ->   category_id VARCHAR(10) NOT NULL,
  ->   title VARCHAR(150) NOT NULL,
  ->   description TEXT,
  ->   budget_min DECIMAL(10,2) CHECK(budget_min > 0),
  ->   budget_max DECIMAL(10,2) CHECK(budget_max > 0),
  ->   job_type VARCHAR(20) CHECK(job_type IN ('Fixed','Hourly')),
  ->   status VARCHAR(20) DEFAULT 'Open',
  ->   posted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  ->   deadline DATE,
  ->
  ->   FOREIGN KEY (client_id)
  ->   REFERENCES CLIENT(client_id),
  ->
  ->   FOREIGN KEY (category_id)
  ->   REFERENCES CATEGORY(category_id)
  -> );
Query OK, 0 rows affected (0.03 sec)

```

```

mysql> CREATE TABLE PROPOSAL (
  -> proposal_id VARCHAR(10) PRIMARY KEY,
  -> job_id VARCHAR(10) NOT NULL,
  -> freelancer_id VARCHAR(10) NOT NULL,
  -> cover_letter TEXT,
  -> bid_amount DECIMAL(10,2) CHECK(bid_amount > 0),
  -> estimated_days INT CHECK(estimated_days > 0),
  -> status VARCHAR(20) DEFAULT 'Pending',
  -> FOREIGN KEY (job_id)
  -> REFERENCES JOB_POST(job_id),
  ->
  -> FOREIGN KEY (freelancer_id)
  -> REFERENCES FREELANCER(freelancer_id)
  -> );CREATE TABLE PROPOSAL (
Query OK, 0 rows affected (0.02 sec)

```

```

mysql> CREATE TABLE CONTRACT (
  -> contract_id VARCHAR(10) PRIMARY KEY,
  -> proposal_id VARCHAR(10) UNIQUE,
  -> client_id VARCHAR(10),
  -> freelancer_id VARCHAR(10),
  -> total_amount DECIMAL(10,2) CHECK(total_amount > 0),
  -> payment_type VARCHAR(20),
  -> status VARCHAR(20),
  -> FOREIGN KEY (proposal_id)
  -> REFERENCES PROPOSAL(proposal_id),
  ->
  -> FOREIGN KEY (client_id)
  -> REFERENCES CLIENT(client_id),
  ->
  -> FOREIGN KEY (freelancer_id)
  -> REFERENCES FREELANCER(freelancer_id)
  -> )
[
  -> ;
Query OK, 0 rows affected (0.02 sec)

```

```
mysql> CREATE TABLE FREELANCER_SKILL (  
  -> freelancer_skill_id VARCHAR(10) PRIMARY KEY,  
  -> freelancer_id VARCHAR(10),  
  -> skill_id VARCHAR(10),  
  -> proficiency_level VARCHAR(20),  
  -> FOREIGN KEY (freelancer_id)  
  -> REFERENCES FREELANCER(freelancer_id)  
  -> ON DELETE CASCADE  
  -> ON UPDATE CASCADE,  
  ->  
  -> FOREIGN KEY (skill_id)  
  -> REFERENCES SKILL(skill_id)  
  -> ON DELETE CASCADE  
  -> ON UPDATE CASCADE  
[ -> );  
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE REVIEW (  
  -> review_id VARCHAR(10) PRIMARY KEY,  
  -> contract_id VARCHAR(10),  
  -> rating INT CHECK(rating BETWEEN 1 AND 5),  
  -> comment TEXT,  
  -> FOREIGN KEY (contract_id)  
  -> REFERENCES CONTRACT(contract_id)  
[ -> );  
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> CREATE TABLE PAYMENT (  
  -> payment_id VARCHAR(10) PRIMARY KEY,  
  -> contract_id VARCHAR(10),  
  -> amount DECIMAL(10,2) CHECK(amount > 0),  
  -> currency VARCHAR(10) DEFAULT 'USD',  
  -> status VARCHAR(20),  
  -> FOREIGN KEY (contract_id)  
  -> REFERENCES CONTRACT(contract_id)  
[ -> );  
Query OK, 0 rows affected (0.02 sec)
```

Question 2

```
mysql> INSERT INTO SKILL
      -> VALUES
[      -> ('S1','PHP','Backend Development');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO CLIENT
      -> VALUES
[      -> ('C1','Ali Khan','ali@gmail.com','111111','Pakistan','ABC Ltd',4.5,NOW());
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO CLIENT
      -> VALUES
[      -> ('C1','Ali Khan','ali@gmail.com','111111','Pakistan','ABC Ltd',4.5,NOW());
ERROR 1062 (23000): Duplicate entry 'C1' for key 'client.PRIMARY'
mysql> INSERT INTO REVIEW
      -> VALUES
[      -> ('R1','CT1',7,'Excellent');
ERROR 3819 (HY000): Check constraint 'review_chk_1' is violated.
mysql> INSERT INTO CLIENT
      -> (client_id,email,country)
      -> VALUES
[      -> ('C3','test@gmail.com','Pakistan');
ERROR 1364 (HY000): Field 'full_name' doesn't have a default value
mysql> INSERT INTO JOB_POST
      -> VALUES
[      -> ('J1','INVALID','CAT1',
      -> 'Website Project',
      -> 'Build Website',
      -> 500,
      -> 1000,
      -> 'Fixed',
      -> 'Open',
      -> NOW(),
      -> '2026-01-01');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`freelance_marketplace`.`job_post`,
CONSTRAINT `job_post_ibfk_1` FOREIGN KEY (`client_id`) REFERENCES `client` (`client_id`))
mysql>
```

All the constraints are working perfectly.

Question 3

```
mysql> ALTER TABLE CLIENT
[      -> ADD CONSTRAINT uq_client_phone UNIQUE(phone);
Query OK, 0 rows affected, 1 warning (0.04 sec)
Records: 0 Duplicates: 0 Warnings: 1
```

```
mysql> ALTER TABLE PAYMENT
      -> ADD CONSTRAINT chk_amount_positive
[      -> CHECK(amount > 0);
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```


Question 4

1. CLIENT
 - PK = client_id
 - UNIQUE = email, phone
 - NOT NULL = full_name, email, country
 - CHECK = avg_rating
 - DEFAULT = created_at
2. FREELANCER
 - PK, UNIQUE, NOT NULL, CHECK, DEFAULT
3. JOB_POST
 - PK, FK, CHECK, DEFAULT
4. PROPOSAL
 - PK, FK, CHECK
5. CONTRACT
 - PK, FK, UNIQUE
6. REVIEW
 - PK, FK, CHECK
7. PAYMENT
 - PK, FK, CHECK, DEFAULT

Question 5

```
mysql> ALTER TABLE CLIENT
[ -> DROP CONSTRAINT uq_client_phone;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

Question 6

```
mysql> INSERT INTO FREELANCER
      -> VALUES
      -> ('F1',
      -> 'John',
      -> 'john@gmail.com',
      -> '12345',
      -> 'USA',
      -> 'Developer',
      -> 30,
      -> 4.8,
      -> 10,
[      -> NOW());
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO SKILL
      -> VALUES
[      -> ('S1','PHP','Backend Development');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO FREELANCER_SKILL
      -> VALUES
      -> ('FS1',
      -> 'F1',
      -> 'S1',
[      -> 'Expert');
Query OK, 1 row affected (0.00 sec)
```

```
[mysql> SELECT * FROM FREELANCER_SKILL;
+-----+-----+-----+-----+
| freelancer_skill_id | freelancer_id | skill_id | proficiency_level |
+-----+-----+-----+-----+
| FS1                 | F1            | S1       | Expert            |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> UPDATE FREELANCER
    -> SET freelancer_id='F100'
[    -> WHERE freelancer_id='F1';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0  Changed: 0  Warnings: 0

[mysql> SELECT * FROM FREELANCER_SKILL;
+-----+-----+-----+-----+
| freelancer_skill_id | freelancer_id | skill_id | proficiency_level |
+-----+-----+-----+-----+
| FS1                 | F100         | S1       | Expert            |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

Child Table updated automatically

```
mysql> DELETE FROM FREELANCER
[    -> WHERE freelancer_id='F100';
Query OK, 0 rows affected (0.00 sec)

[mysql> SELECT * FROM FREELANCER_SKILL;
Empty set (0.00 sec)
```

Question 7

```
[mysql> SELECT * FROM PAYMENT;
Empty set (0.00 sec)

[mysql> TRUNCATE TABLE PAYMENT;
Query OK, 0 rows affected (0.02 sec)
```

```
[mysql> SELECT * FROM PAYMENT;
Empty set (0.00 sec)

[mysql> DROP TABLE REVIEW;
Query OK, 0 rows affected (0.01 sec)
```

```
[mysql> SELECT * FROM REVIEW;
ERROR 1146 (42S02): Table 'freelance_marketplace.review' doesn't exist
```

